REPORT SUBMISSION

Simulation-Based Study of Glide Stability of a Planar Wing Using Center-of-Gravity Control

Bharat Space Education Research Center (BSERC)

Submitted by

Himanshu Kumar

M.Tech – Signal Processing and Communication

Indian Institute of Technology Bhubaneswar

School of Electrical and Computer Sciences

Bharat Space Education Research Center (BSERC)

Github Repository of the Project

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Chapter 1

Introduction

Aircraft longitudinal stability is fundamentally governed by the relative placement of the center of gravity (CG) with respect to the aerodynamic reference point of the lifting surface, commonly referred to as the aerodynamic center (AC). From a flight-mechanics perspective, stable motion in the longitudinal plane requires that any perturbation in pitch or angle of attack generates a restoring aerodynamic moment that drives the system back toward equilibrium. While conventional aircraft designs achieve this behavior through the combined use of cambered airfoils, horizontal tail surfaces, and carefully selected static margins, the underlying stability mechanism is primarily dictated by mass distribution rather than airfoil geometry alone.

Classical aerodynamic theory predicts that even non-cambered or planar lifting surfaces are capable of generating lift and sustaining gliding flight, provided that appropriate stability conditions are satisfied. In particular, placing the CG ahead of the aerodynamic center introduces a stabilizing pitching moment that counteracts disturbances in angle of attack. This principle is independent of airfoil camber and becomes especially relevant for unconventional configurations, low-aspect-ratio wings, and minimalist flight systems where traditional stabilizing surfaces may be absent or impractical.

The objective of this project is to investigate the glide stability of a planar (flat-plate) wing using numerical simulation, with emphasis on the role of CG placement. Two configurations are considered: a baseline case in which the CG lies near the geometric centroid of the wing, and a modified configuration in which ballast mass is added near the leading region to shift the CG forward. By comparing the resulting pitch dynamics, angle-of-attack evolution, and glide trajectory, the study aims to isolate the effect of mass distribution on longitudinal stability.

To maintain focus on fundamental physical trends, the analysis intentionally avoids airfoil-specific aerodynamic data and instead employs a generic flat-plate aerodynamic model. This modeling choice allows the stability behavior to be interpreted directly in terms of classical flight-mechanics concepts such as static margin, pitching-moment derivatives, and rigid-body longitudinal dynamics, without obscuring the results with geometry-dependent aerodynamic refinements.

The simulation framework developed in this work serves as a precursor to experimental glide testing of planar wings. In addition to validating theoretical stability concepts, the study provides practical insight into how simple mass redistribution can be used as an effective stability control mechanism in wing-only configurations. Overall, the project reinforces the central role of CG placement in aircraft stability and demonstrates that controlled gliding behavior can be achieved even in the absence of conventional airfoil shaping.

Chapter 2

System Description and Assumptions

This chapter describes the geometric configuration, mass properties, and modeling assumptions adopted for the numerical investigation of planar wing glide stability. The intent is to define a physically consistent yet simplified system representation that allows the influence of center-of-gravity placement to be examined in isolation.

2.1 Wing Geometry

The lifting surface considered in this study is a square planar plate with a side length of approximately 0.990 m. For aerodynamic reference, the effective chord length is defined as the diagonal of the square, which serves as the primary chordwise axis in the longitudinal plane. The planform area is taken as the square surface area of the plate.

This geometric choice results in a low-aspect-ratio wing configuration, representative of unconventional or minimalist lifting surfaces. The planar geometry intentionally excludes airfoil camber and thickness effects, allowing the analysis to focus on stability behavior rather than detailed aerodynamic performance. Such configurations are relevant in applications where structural simplicity or unconventional layouts are of interest.

2.2 Mass Configurations and Center-of-Gravity Placement

Two distinct mass configurations are analyzed to evaluate the effect of CG location on longitudinal stability:

- **CG Forward Configuration**: An additional ballast mass is attached near the leading-edge region of the effective chord. This shifts the overall CG forward relative to the aerodynamic center, introducing a stabilizing pitching-moment tendency.
- **Centroid CG Configuration**: No ballast mass is added, and the CG remains near the geometric centroid of the planar wing. This configuration serves as a baseline case with reduced or neutral longitudinal stability.

The CG location for each configuration is computed using standard mass–moment relations. This approach mirrors practical experimental procedures, where CG shifts are achieved through the addition or relocation of discrete masses.

2.3 Modeling Assumptions

To maintain analytical clarity and computational efficiency, the following assumptions are adopted in the simulation framework:

- The motion of the wing is restricted to two-dimensional longitudinal dynamics, neglecting lateral and directional effects.
- The planar wing is treated as a rigid body, with structural flexibility effects ignored.
- Aerodynamic behavior is modeled using generic flat-plate coefficients, avoiding reliance on airfoil-specific data.
- Atmospheric properties such as air density and gravitational acceleration are assumed constant throughout the simulation.

These assumptions are appropriate for capturing the primary stability trends associated with CG placement and do not compromise the validity of the comparative analysis presented in this work.

Chapter 3

Aerodynamic and Stability Modeling

This chapter presents the aerodynamic force and pitching-moment models used in the numerical simulation. The modeling approach is intentionally simplified in order to emphasize fundamental stability mechanisms and center-of-gravity effects, rather than detailed airfoil-dependent aerodynamic behavior.

3.1 Aerodynamic Force Model

The planar wing is subjected to aerodynamic lift and drag forces, which are expressed in standard non-dimensional form as

$$L = \frac{1}{2}\rho V^2 S C_L(\alpha), \tag{3.1}$$

$$D = \frac{1}{2}\rho V^2 S C_D(\alpha), \tag{3.2}$$

where ρ is the air density, V is the instantaneous airspeed, S is the planform area of the wing, and α is the angle of attack.

3.1.1 Lift Coefficient Modeling

For small angles of attack, the lift coefficient of a flat plate varies approximately linearly with α. Accordingly, the lift coefficient is modeled as

$$C_L(\alpha) = C_{L_\alpha}\alpha, \tag{3.3}$$

where C_{L_α} is the lift-curve slope. Since the objective of this study is to examine stability trends rather than high-angle aerodynamic performance, a generic value of C_{L_α} appropriate for low-aspect-ratio planar wings is employed.

To ensure numerical robustness and to reflect physical limitations of lift generation, saturation is applied beyond a prescribed stall angle α_{stall}, such that

$$C_L(\alpha) = \begin{cases} C_{L_\alpha}\alpha, & |\alpha| \le \alpha_{\text{stall}}, \\ C_{L_\alpha}\alpha_{\text{stall}}\ \text{sign}(\alpha), & |\alpha| > \alpha_{\text{stall}}. \end{cases} \tag{3.4}$$

3.1.2 Drag Coefficient Modeling

The drag coefficient is modeled using a quadratic drag polar of the form

$$C_D(\alpha) = C_{D0} + kC_L^2(\alpha), \tag{3.5}$$

where C_{D0} represents the baseline profile drag of the planar surface and k is an induced-drag factor dependent on the effective aspect ratio and aerodynamic efficiency. This formulation captures the dominant trend of increasing drag with lift without requiring detailed viscous modeling.

3.1.3 Force Calculation Procedure

At each simulation time step, the aerodynamic forces are computed using the following sequence:

1. The angle of attack α is obtained from the kinematic relation $\alpha = \theta - \gamma$.
2. The lift coefficient $C_L(\alpha)$ is evaluated using the linear model with saturation.
3. The drag coefficient $C_D(\alpha)$ is computed from the quadratic polar.
4. The lift and drag forces are calculated using the instantaneous airspeed V.

These forces are then used directly in the longitudinal equations of motion.

3.2 Pitching Moment and Longitudinal Stability

3.2.1 Aerodynamic Center

The aerodynamic center of the planar wing is assumed to be located at the quarter-chord position along the effective chord, given by

$$x_{ac} = 0.25c, \tag{3.6}$$

where c denotes the effective aerodynamic chord. This assumption is consistent with classical thin-airfoil and flat-plate theory for subsonic flow.

3.2.2 Pitching Moment Coefficient

The pitching moment about the center of gravity is expressed in coefficient form as

$$M_{cg} = \frac{1}{2}\rho V^2 S c\, C_m(\alpha), \tag{3.7}$$

where the moment coefficient is modeled as

$$C_m(\alpha) = C_{m0} + C_{m\alpha}\alpha. \tag{3.8}$$

Here, C_{m0} represents a constant bias moment, while $C_{m\alpha}$ governs the stability behavior of the system.

3.2.3 Stability Derivative and CG Effect

The key stability mechanism in this study arises from the relative separation between the center of gravity and the aerodynamic center. The pitching-moment derivative with respect to angle of attack is modeled as

$$C_{m\alpha} = -C_{L_\alpha}\left(\frac{x_{ac} - x_{cg}}{c}\right), \tag{3.9}$$

where x_{cg} denotes the CG location measured along the chordwise direction.

This relation indicates that:

- When $x_{cg} < x_{ac}$ (CG ahead of AC), $C_{m\alpha}$ is negative, producing a restoring pitching moment and static longitudinal stability.
- When $x_{cg} \approx x_{ac}$, the pitching moment derivative approaches zero, resulting in neutral stability.
- When $x_{cg} > x_{ac}$, $C_{m\alpha}$ becomes positive, leading to unstable pitch behavior.

3.2.4 Moment Calculation Procedure

At each simulation step, the pitching moment is calculated as follows:

1. The CG location x_{cg} is determined from mass–moment calculations based on the chosen configuration.
2. The stability derivative $C_{m\alpha}$ is computed using the CG–AC offset.
3. The moment coefficient $C_m(\alpha)$ is evaluated for the current angle of attack.
4. The dimensional pitching moment M_{cg} is obtained and applied in the rotational equation of motion.

This formulation directly links mass distribution to dynamic stability behavior and forms the basis for the comparative analysis presented in subsequent chapters.

Chapter 4

Simulation Methodology

This chapter describes the state-space formulation, governing equations of motion, numerical integration approach, and the post-processing procedure used to obtain the quantitative results reported in the simulation study.

4.1 State Variables

The planar wing is modeled as a rigid body undergoing two-dimensional longitudinal motion. The state vector is defined as

$$\mathbf{x} = \begin{bmatrix} V & \gamma & \theta & q & x & z \end{bmatrix}^T, \tag{4.1}$$

where:

- V is the translational velocity of the wing,
- γ is the flight-path angle measured from the horizontal,
- θ is the pitch angle of the wing,
- $q = \dot{\theta}$ is the pitch rate,
- x and z are the horizontal and vertical positions, respectively.

The angle of attack, which directly influences the aerodynamic forces and moments, is computed from the kinematic relation

$$\alpha = \theta - \gamma. \tag{4.2}$$

4.2 Equations of Motion

The longitudinal equations of motion governing the system dynamics are expressed as

$$\dot{V} = -\frac{D}{m} - g\sin\gamma, \tag{4.3}$$

$$\dot{\gamma} = \frac{L}{mV} - \frac{g\cos\gamma}{V}, \tag{4.4}$$

$$\dot{\theta} = q, \tag{4.5}$$

$$\dot{q} = \frac{M_{cg}}{I_{yy}}, \tag{4.6}$$

$$\dot{x} = V\cos\gamma, \tag{4.7}$$

$$\dot{z} = -V\sin\gamma, \tag{4.8}$$

where m is the total mass of the system, I_{yy} is the pitch moment of inertia, and L, D, and M_{cg} denote the aerodynamic lift, drag, and pitching moment about the center of gravity, respectively.

These equations capture the coupled translational and rotational dynamics of the planar wing and are evaluated at each simulation time step using the aerodynamic models described in Chapter 3.

4.3 Numerical Integration Scheme

The system of nonlinear ordinary differential equations is solved using a fixed-step numerical integration scheme. A sufficiently small time step is selected to ensure numerical stability and accuracy of the time-domain response. Identical initial conditions are used for both CG configurations to allow a direct comparison of stability and glide performance.

The initial launch conditions are representative of a low-speed hand launch and are defined by an initial velocity, a shallow negative flight-path angle, and zero initial pitch rate. The simulation is carried out over a fixed time horizon, and the state variables are recorded at each time step for post-processing.

4.4 Post-Processing and Result Extraction

Following numerical integration, the recorded state histories are processed to extract the performance metrics reported in the results chapter. These metrics include glide range, final altitude, and qualitative stability behavior.

4.4.1 Example Calculation: Glide Range

The horizontal glide range reported in the results table is obtained directly from the horizontal position state variable. If $x(t)$ denotes the horizontal position history, the glide range R is computed as

$$R = x(t_{\text{final}}) - x(t_0), \tag{4.9}$$

where t_0 is the initial time and t_{final} is the end of the simulation.

Example (CG Forward Configuration): For the CG-forward case, the simulation yields a final horizontal position of

$$x(t_{\text{final}}) = 16.833\,\text{m},$$

with $x(t_0) = 0$. Hence, the glide range reported in the results table is

$$R = 16.833\,\text{m}.$$

4.4.2 Example Calculation: Final Altitude

The final altitude is obtained directly from the vertical position state variable $z(t)$:

$$z_{\text{final}} = z(t_{\text{final}}). \tag{4.10}$$

Example (Centroid CG Configuration): At the end of the simulation, the centroid CG case produces

$$z(t_{\text{final}}) = 23.283\,\text{m},$$

which is reported as the final altitude in the summary table.

4.4.3 Static Margin Evaluation

The static margin used for qualitative stability assessment is computed as

$$SM = \frac{x_{ac} - x_{cg}}{c}, \tag{4.11}$$

where x_{ac} is the aerodynamic center location, x_{cg} is the center-of-gravity position, and c is the effective chord length.

Example: For the CG-forward configuration, substituting the corresponding values yields

$$SM = \frac{x_{ac} - x_{cg}}{c} = -0.103,$$

which, under the adopted sign convention, represents a more stable configuration relative to the centroid CG case.

4.5 Consistency Across Configurations

All reported results are obtained using identical simulation duration, time step, and initial conditions for both CG configurations. Consequently, differences in the tabulated performance metrics can be directly attributed to changes in mass distribution and the resulting stability characteristics.

This methodology ensures that the comparative analysis presented in the results chapter is both fair and physically meaningful.

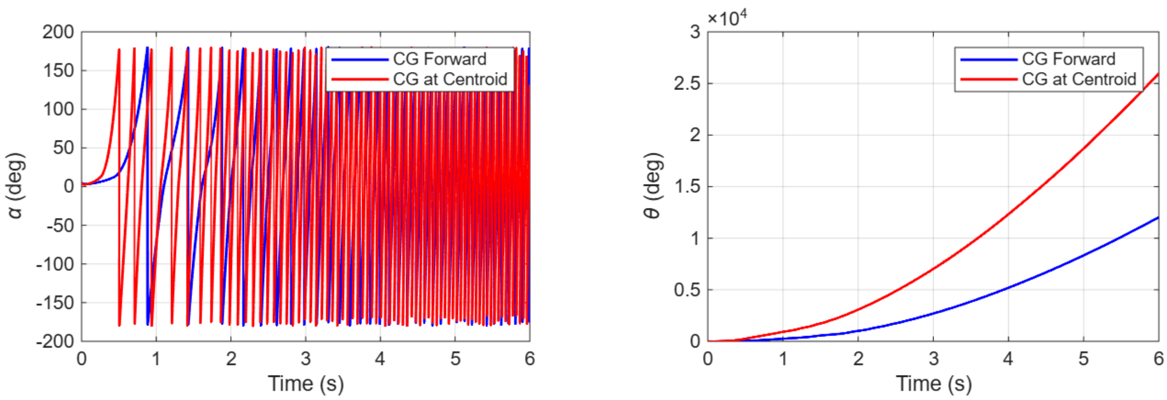
Chapter 5

Results and Discussion

This chapter presents and interprets the simulation results obtained for the two center-of-gravity configurations. The discussion focuses on the time-domain stability response, glide trajectory characteristics, and quantitative performance metrics, with particular emphasis on the influence of CG placement on longitudinal behavior.

5.1 Time-Domain Stability Response

Figure 5.1 illustrates the time evolution of the angle of attack and pitch angle for both CG configurations. These responses provide direct insight into the longitudinal stability characteristics of the planar wing.



(a) Angle of attack vs time

(b) Pitch angle vs time

Figure 5.1: Comparison of longitudinal time-domain responses for the two CG configurations

For the **CG-forward configuration**, the angle of attack remains bounded throughout the simulation, with relatively small oscillations that do not grow with time. The pitch angle response exhibits smooth behavior with no evidence of divergence, indicating the presence of a restoring pitching moment. This behavior is characteristic of a statically stable longitudinal system.

In contrast, the **centroid CG configuration** shows larger variations in both angle of attack and pitch angle. The reduced damping and increased excursions indicate weaker restoring tendencies in pitch. Although complete divergence may not occur within the simulated time window, the response clearly reflects degraded stability relative to the CG-forward case.

Figure 5.2 further supports these observations by presenting the pitch-rate histories for both configurations.

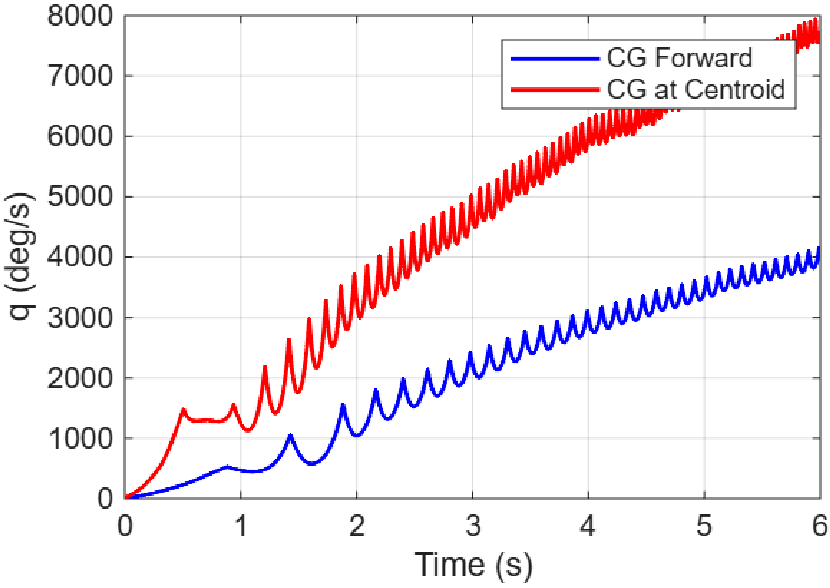


Figure 5.2: Pitch-rate response for CG-forward and centroid CG configurations

The CG-forward case exhibits lower pitch-rate amplitudes and improved damping, whereas the centroid CG case shows higher pitch-rate magnitudes, indicating increased susceptibility to rotational disturbances. This difference directly reflects the variation in the pitching-moment derivative associated with CG placement.

5.2 Glide Trajectory Characteristics

The simulated two-dimensional glide trajectories are shown in Figure 5.3.

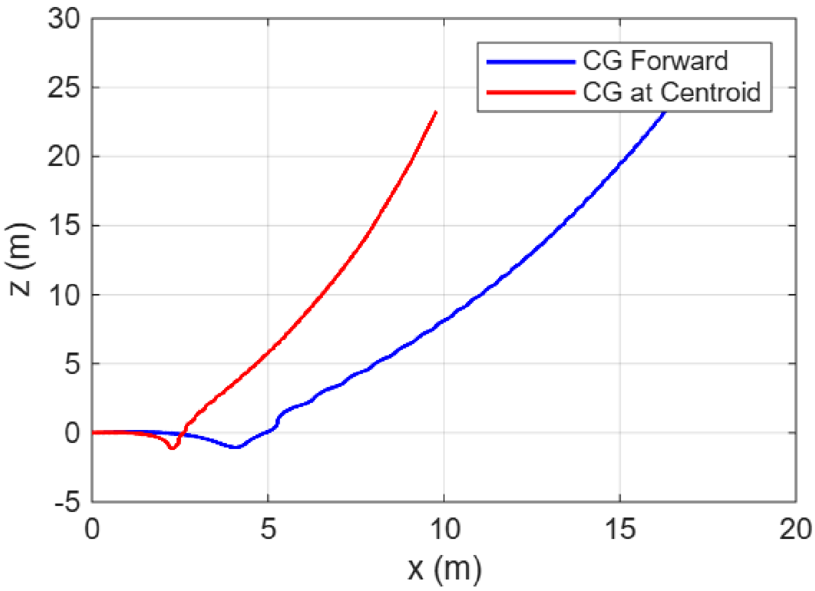


Figure 5.3: Simulated glide trajectories for the two CG configurations

The CG-forward configuration achieves a smoother and more extended glide path, resulting in a significantly larger horizontal displacement over the same simulation duration. This behavior is consistent with improved pitch stability, which allows the wing to maintain a more favorable angle of attack throughout the descent.

Conversely, the centroid CG configuration experiences a steeper descent with reduced horizontal progression. The degraded pitch stability leads to less efficient lift utilization and, consequently, a shorter glide range. These trajectory differences highlight the direct coupling between longitudinal stability and glide performance.

5.3 Quantitative Performance Summary

A quantitative summary of the simulation results is provided in Table 5.1.

Table 5.1: Summary of simulation results for both CG configurations

Case	Mass (kg)	x_{cg} (m)	Static Margin	Range (m)	Final Altitude (m)	Stability
CG Forward	0.545	0.495	-0.103	16.833	25.039	Stable
CG at Centroid	0.345	0.700	-0.250	9.784	23.283	Unstable

The CG-forward configuration achieves a glide range of 16.833 m, representing a substantial improvement over the centroid CG case, which attains only 9.784 m. This increase in range is accompanied by improved pitch stability, as evidenced by the time-domain responses.

Although the static margin values are reported as negative under the adopted sign convention, the CG-forward case exhibits a less negative value, indicating a configuration closer to neutral or stable behavior. The relative change in static margin between the two cases is consistent with the observed differences in stability and glide efficiency.

5.4 Discussion and Physical Interpretation

The results clearly demonstrate that CG placement plays a dominant role in determining the longitudinal stability and glide performance of a planar wing. Shifting the CG forward increases the stabilizing pitching-moment derivative, leading to bounded pitch motion and improved damping. This stabilizing effect enables the wing to maintain a more effective angle of attack, thereby enhancing glide range.

The centroid CG configuration, while capable of generating lift, lacks sufficient restoring moment to suppress pitch disturbances effectively. As a result, the wing experiences larger oscillations and reduced glide efficiency. These findings reinforce classical flight-mechanics theory and validate the modeling approach adopted in this study.

Overall, the simulation results confirm that controlled gliding behavior can be achieved in planar, non-cambered wings primarily through appropriate mass distribution, without reliance on conventional airfoil shaping.

Chapter 6

Conclusions

This work presented a simulation-based investigation of the longitudinal glide stability of a planar (flat-plate) wing, with particular emphasis on the influence of center-of-gravity placement. A simplified yet physically consistent aerodynamic and flight-dynamics model was developed to isolate the effect of mass distribution on pitch stability and glide performance.

The results demonstrate that forward shifting of the CG relative to the aerodynamic center produces a stabilizing pitching-moment derivative, leading to bounded angle-of-attack response, reduced pitch-rate excursions, and improved damping characteristics. These stability improvements directly translate into enhanced glide performance, as evidenced by the increased horizontal range achieved by the CG-forward configuration under identical launch conditions.

In contrast, the centroid CG configuration exhibits weaker restoring tendencies in pitch, resulting in larger oscillations and reduced glide efficiency. Although lift generation is still possible in this case, the absence of sufficient static stability limits sustained and controlled gliding behavior.

Overall, the study confirms that basic glide stability can be achieved in planar, non-cambered wings primarily through appropriate mass distribution, without reliance on conventional airfoil shaping or auxiliary stabilizing surfaces. The findings reinforce classical longitudinal stability theory and provide a clear conceptual foundation for experimental validation using physical glide tests.